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CPDA: Class-Conditional Path Distribution Alignment for Unsupervised Time-Series Domain Adaptation

A Composite Signature-Spectral Kernel Beats 30 Baselines on 13 Time-Series DA Benchmarks Without Adversarial Training

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Most unsupervised time-series domain adaptation (UTS-DA) methods align only global marginal feature distributions, discarding the class structure that defines decision boundaries. CPDA instead aligns class-conditional latent *path distributions* via a novel composite signature-spectral kernel that jointly captures temporal trajectory structure, frequency content, and semantic features. Evaluated on 13 benchmarks against 30 baselines, CPDA achieves consistent state-of-the-art results with CNN, ResNet18, and TCN backbones—without adversarial training.

AT A GLANCE

Problem: UTS-DA distribution shifts arise from different users, sensors, or temporal dynamics; prior methods align only global marginals, losing class structure.

Why it matters: Labeling data for every new deployment context is infeasible; UTS-DA enables transfer without target labels in IoT, health, and manufacturing.

Method: Class-conditional alignment via a composite kernel combining path signatures, spectral features, semantic pooling, and low-rank dynamics.

Result: State-of-the-art on 13 benchmarks against 30 baselines with 3 backbone architectures.

Evidence: Non-adversarial; stable training across all experimental configurations.

The Big Picture

Time-series classifiers deployed in the real world face a pervasive problem: the distribution of training data differs from the distribution at deployment. A human activity recognition model trained on data from a 25-year-old male with one wearable device may perform poorly on a 60-year-old female with a different device. A machine fault-detection system trained on one production line may fail on a nominally identical line with slightly different resonance characteristics.

Unsupervised time-series domain adaptation addresses this by transferring knowledge from a labeled **source domain** to an unlabeled **target domain**. The key insight of CPDA is that temporal data has rich structure—**path trajectories** in feature space, **frequency spectra**, and **class-specific distributional signatures**—that global marginal alignment discards. CPDA preserves this structure through a principled kernel design.

Why Existing Approaches Fall Short

The dominant paradigm in UTS-DA follows the deep domain adaptation tradition from computer vision: minimize a distributional discrepancy (e.g., MMD) or train a domain adversary (e.g., DANN) on marginal feature distributions. Both approaches have structural limitations for time-series data.

Global marginal alignment ignores class structure. If source class A aligns with target class B due to superficial distributional similarity, a classifier that looked correct in source space will fail in target space. This **negative transfer** effect is well-known but remains underaddressed in UTS-DA.

Spectral and temporal structure is discarded when features are pooled before alignment. A model that misses the difference between a 10 Hz and a 15 Hz vibration pattern may align the two classes' pooled statistics while failing on the actual classification problem.

Adversarial training is unstable, sensitive to hyperparameters, and prone to mode collapse, making it difficult to deploy reliably across diverse benchmark settings.

Core Mechanism

CPDA operates in the encoder's latent space, treating each sample as a **path**—a sequence of latent feature vectors over time. It aligns class-conditional path distributions via a composite kernel k_{CPDA} that combines four components:

1. **Semantic pooling kernel:** Classic RBF kernel on time-averaged latent features; captures overall distributional similarity.
2. **Path-signature kernel:** Measures similarity of ordered polynomial statistics of the time-series trajectory—captures temporal dynamics.
3. **Spectral kernel:** Fourier-based kernel on the frequency-domain representation of the latent path—captures periodic structure.
4. **Low-rank path-signature kernel:** Compressed, dominant modes of temporal variation via truncated signature expansion—reduces noise sensitivity.

Technical Formulation

Let $\phi_\theta : \mathcal{X} \to \mathbb{R}^{T\times d}$ be the encoder mapping a time-series x to a T-step latent path. Define the class-conditional source and target distributions:

$$P_S^y = \{\phi_\theta(x) \mid x \in \mathcal{D}_S, y_x = y\}, \quad P_T^y = \{\phi_\theta(x) \mid x \in \mathcal{D}_T, \hat{y}_x = y\}$$

where $\hat{y}_x$ are pseudo-labels for target samples.

The CPDA discrepancy is:

$$\mathcal{D}_{\text{CPDA}} = \sum_{y=1}^{C} \text{MMD}^2_{k_{\text{CPDA}}}(P_S^y, P_T^y)$$

with:

$$k_{\text{CPDA}}(\gamma,\gamma') = \alpha_1 k_{\text{sem}}(\gamma,\gamma') + \alpha_2 k_{\text{sig}}(\gamma,\gamma') + \alpha_3 k_{\text{spec}}(\gamma,\gamma') + \alpha_4 k_{\text{lr}}(\gamma,\gamma')$$

where γ, γ' are latent paths and α_i are learned mixture weights.

The training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(\mathcal{D}_S) + \lambda\, \mathcal{D}_{\text{CPDA}}(\mathcal{D}_S, \mathcal{D}_T)$$

Learning or Inference Procedure

Pseudo-label generation: Target pseudo-labels are produced by the source-trained classifier and updated iteratively as the encoder improves. High-confidence pseudo-labels (above a threshold) are used for alignment.

Optimization: Stochastic gradient descent on $\mathcal{L}$; no adversarial loop, no discriminator. The MMD-based objective is smooth and differentiable.

Inference: Standard forward pass through the adapted encoder and classifier; no test-time computation overhead relative to the base model.

What the Guarantee Says

Under standard domain adaptation assumptions (covariate shift, shared label conditional), minimizing class-conditional MMD drives the target encoder toward the source class-conditional distributions, providing a theoretical basis for classifier transfer. The path-signature kernel is characteristic on path space under mild regularity conditions, guaranteeing that $\mathcal{D}_{\text{CPDA}} = 0$ implies equality of path distributions. In practice, finite-sample approximation introduces error, but the rich kernel reduces the approximation gap relative to simpler kernels.

Experimental Findings

- **13 benchmarks:** Including activity recognition (HAR, UCIHAR, WISDM), fault detection (CWRU, PU), and physiological monitoring (EEG, ECG).
- **30 baselines:** Spanning discrepancy-based (MMD, CORAL), adversarial (DANN, CDAN, AdvSKM), and pseudo-labeling methods (SHOT, SDAT).
- **3 backbones:** CNN, ResNet18, TCN—CPDA achieves state-of-the-art with all three.
- **Margin:** CPDA outperforms the second-best method on average across all 13 benchmarks by a consistent margin with all backbone choices.

Ablations and Interpretation

Kernel component ablation. Removing any single component (k_{sig}, k_{spec}, k_{lr}, k_{sem}) degrades performance on most benchmarks, confirming that each component captures complementary distributional information. The path-signature kernel has the largest marginal contribution on benchmarks with strong temporal dynamics (fault detection); the spectral kernel dominates on benchmarks with periodic signals (EEG, ECG).

Global vs. class-conditional alignment. Replacing class-conditional alignment with global marginal MMD using k_{CPDA} degrades performance substantially on benchmarks with closely adjacent class clusters, confirming the negative-transfer hypothesis.

Non-adversarial vs. adversarial. CPDA trains stably across all 13 benchmarks; GAN-based methods (DANN, CDAN) show higher variance and failed convergence on 3 of 13 benchmarks.

Reference

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